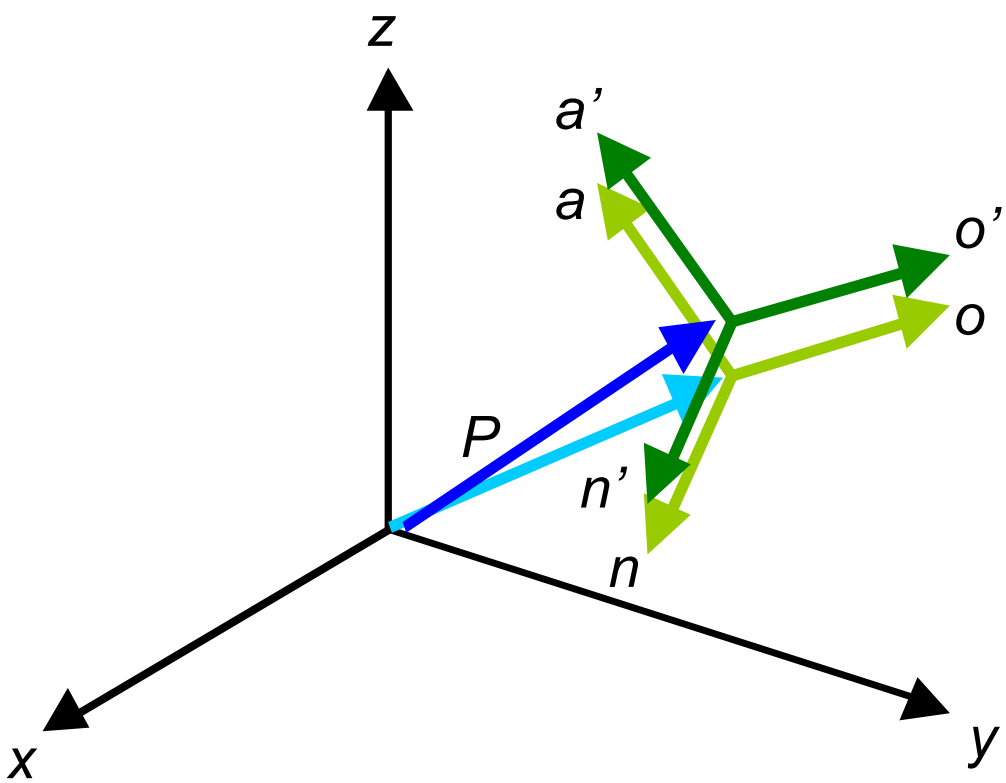
Differential Kinematics

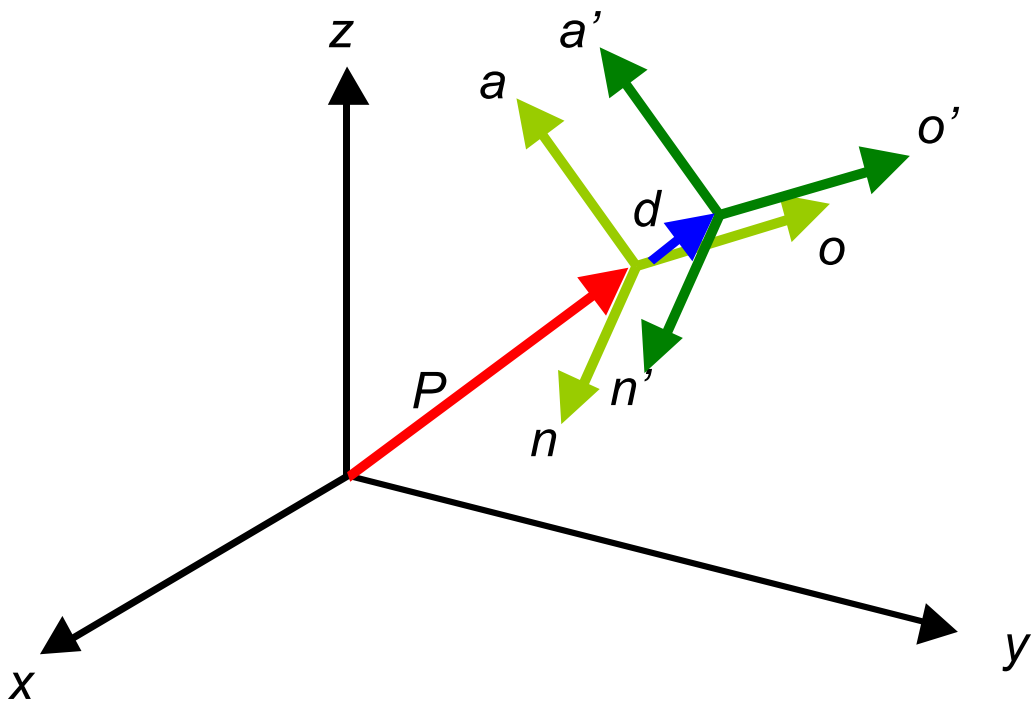
Differential Kinematics describe small motions of a reference frame. These differential motions are related to velocity.



Differential Motions

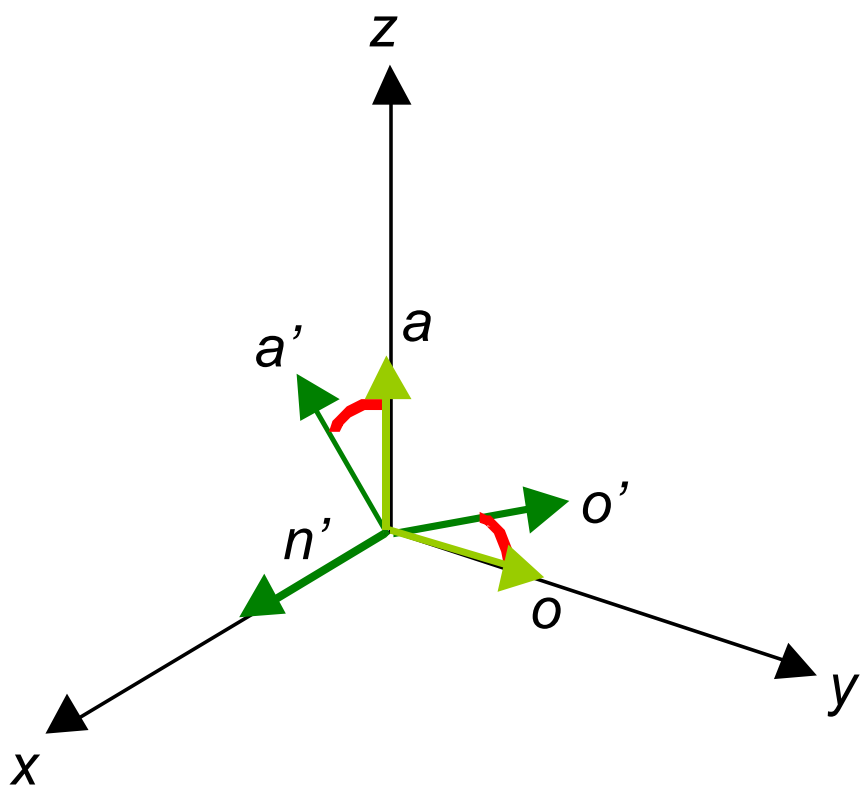
- Differential Translations
- Differential Rotations
- Differential Transformations (combination of Translations and Rotations)

Differential Translation



$$Trans(dx,dy,dz)=\begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Differential Rotation



Since we are using small angles (i.e. δx, δy, δz):

$$\sin \delta x \approx \delta x$$

$$\cos \delta x \approx 1$$

Differential Rotations

Differential Rotation about the x-axis

$$\mathrm{Rot}(x,\delta x)=\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & C\delta x & -S\delta x & 0\\ 0 & S\delta x & C\delta x & 0\\ 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & -\delta x & 0\\ 0 & \delta x & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

Differential Rotation about the y-axis

$$\mathrm{Rot}(y,\delta y)=\begin{bmatrix}C\delta y & 0 & S\delta y & 0\\ 0 & 1 & 0 & 0\\ -S\delta y & 0 & C\delta y & 0\\ 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & \delta y & 0\\ 0 & 1 & 0 & 0\\ -\delta y & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

Differential Rotation about the z-axis

$$\mathrm{Rot}(z,\delta z)=\begin{bmatrix}C\delta z & -S\delta z & 0 & 0\\ S\delta z & C\delta z & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & -\delta z & 0 & 0\\ \delta z & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

Differential Rotation about a general axis

Rotation about a general axis *k* can be represented as sequential rotations about the x, y and z-axes:

$$Rot(k, d\theta) = Rot(x, \delta x)Rot(y, \delta y)Rot(z, \delta z)$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -\delta x & 0 \\ 0 & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \delta y & 0 \\ 0 & 1 & 0 & 0 \\ -\delta y & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\delta z & 0 & 0 \\ \delta z & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -\delta z & \delta y & 0 \\ \delta x \delta y + \delta z & -\delta x \delta y \delta z + 1 & -\delta x & 0 \\ -\delta y + \delta x \delta z & \delta x + \delta y \delta z & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Differential Rotation about a general axis

By using the multiple small angles approximation the rotation about axis *k* can be simplified to:

$$Rot(k,d\theta) = Rot(x,\delta x)Rot(y,\delta y)Rot(z,\delta z)$$

$$= \begin{bmatrix} 1 & -\delta z & \delta y & 0 \\ \delta z & 1 & -\delta x & 0 \\ -\delta y & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Differential Transformation of a Frame

The Differential Transformations of a frame involves the combination of a differential translation followed by a differential rotation about the moving frame. This can be represented as:

$$[T + dT] = [Trans(dx, dy, dz)Rot(k, d\theta)][T]$$

dT is the differential change in the frame *T*. It can be written:

$$[dT] = [\Delta][T] = [Trans(dx, dy, dz)Rot(k, d\theta) - I][T]$$

Differential Operator

The Differential Operator Δ is represented as:

$$\Delta = Trans(dx, dy, dz)Rot(k, d\theta) - I$$

$$= \begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\delta z & \delta y & 0 \\ \delta z & 1 & -\delta x & 0 \\ -\delta y & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -\delta z & \delta y & dx \\ \delta z & 0 & -\delta x & dy \\ -\delta y & \delta x & 0 & dz \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Therefore the differential change *dT* can be written:

$$dT = \begin{bmatrix} dn_x & do_x & da_x & dp_x \\ dn_y & do_y & da_y & dp_y \\ dn_z & do_z & da_z & dp_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -\delta z & \delta y & dx \\ \delta z & 0 & -\delta x & dy \\ -\delta y & \delta x & 0 & dz \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot T$$

Differential Change between Frames

The Differential Operator Δ is represented relative to the fixed frame. To define the Differential Operator for a moving frame (${}^{T}\Delta$) we postmultiple the Frame matrix T:

$$[dT] = [\Delta][T] = [T]\left[{}^{T}\Delta\right]$$

Then premultiple both sides by the inverse of the Frame matrix T:

$$[T]^{-1}[\Delta][T] = [T]^{-1}[T]\left[{}^{T}\Delta\right]$$

$$\left[{}^{T}\Delta\right] = [T]^{-1}[\Delta][T]$$

This gives:

$$\left[{}^{T}\Delta\right] = \left[T\right]^{-1}\left[\Delta\right]\left[T\right]$$

$$= \begin{bmatrix} n_x & n_y & n_z & -p\cdot n \\ o_x & o_y & o_z & -p\cdot o \\ a_x & a_y & a_z & -p\cdot a \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -\delta z & \delta y & dx \\ \delta z & 0 & -\delta x & dy \\ -\delta y & \delta x & 0 & dz \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -{}^{T}\delta z & {}^{T}\delta y & {}^{T}dx \\ {}^{T}\delta z & 0 & -{}^{T}\delta x & {}^{T}dy \\ -{}^{T}\delta y & {}^{T}\delta x & 0 & {}^{T}dz \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

where:

$$
\begin{aligned}
{}^{T}\delta x &= \bar{\delta} \bullet \bar{n} \\
{}^{T}\delta y &= \bar{\delta} \bullet \bar{o} \\
{}^{T}\delta z &= \bar{\delta} \bullet \bar{a} \\
{}^{T}dx &= \bar{n} \bullet \left[\left(\bar{\delta} \times \bar{p} \right) + \bar{d} \right] \\
{}^{T}dy &= \bar{o} \bullet \left[\left(\bar{\delta} \times \bar{p} \right) + \bar{d} \right] \\
{}^{T}dz &= \bar{a} \bullet \left[\left(\bar{\delta} \times \bar{p} \right) + \bar{d} \right]
\end{aligned}
$$